DS_A1_Q5

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1 Q5

To show that test MSE of model $Y = f(x) + \epsilon$,

$$\mathrm{E}[(Y_0 - \hat{f}(x_0))^2] = \mathrm{var}(\hat{f}(x_0)) + b_f(\hat{f}(x_0))^2 + \mathrm{var}(\epsilon)$$

We could write the expression of test MSE first,

$$\begin{aligned}
\mathrm{E}[(Y_0 - \hat{f}(x_0))^2] &= \mathrm{E}[(f(x_0) + \epsilon - \hat{f}(x_0))^2] \\
&= \mathrm{E}[((f(x_0) - \hat{f}(x_0)) + \epsilon)^2] \\
&= \mathrm{E}[(f(x_0) - \hat{f}(x_0)^2] + 2\,\mathrm{E}[(f(x_0) - \hat{f}(x_0) \cdot \mathrm{E}[\epsilon] + \mathrm{E}[\epsilon^2] \\
&= \mathrm{E}[(f(x_0) - \hat{f}(x_0)^2] + 2\,\mathrm{E}[(f(x_0) - \hat{f}(x_0)]\,\mathrm{E}[\epsilon] + \mathrm{E}[\epsilon^2]
\end{aligned}$$

Since ϵ is a random error term that is independent of x and has mean zero, so

$$E[\epsilon] = 0, \quad \mathrm{var}(\epsilon) = \mathrm{E}[\epsilon^2]$$

Substitute the formulas into the derived test MSE expression,

$$\begin{aligned}
\mathrm{E}[(Y_0 - \hat{f}(x_0))^2] &= \mathrm{E}[(f(x_0) - \hat{f}(x_0)^2] + 2\,\mathrm{E}[(f(x_0) - \hat{f}(x_0)] \cdot 0 + \mathrm{var}(\epsilon) \\
&= \mathrm{E}[(f(x_0) - \hat{f}(x_0)^2] + \mathrm{var}(\epsilon)
\end{aligned}$$

To further prove the result, we transform $f(x_0) - \hat{f}(x_0)$ as follows.

$$f(x_0) - \hat{f}(x_0) = (f(x_0) - \mathrm{E}[\hat{f}(x_0)]) + (\mathrm{E}[\hat{f}(x_0)] - \hat{f}(x_0))$$

Substitute to $\mathrm{E}[(f(x_0) - \hat{f}(x_0)^2]$, we can obtain that,

$$\mathrm{E}[(f(x_0)-\hat{f}(x_0))^2] = \mathrm{E}[(\mathrm{E}[\hat{f}(x_0)]-\hat{f}(x_0))^2]+\mathrm{E}[(f(x_0)-E[\hat{f}(x_0)])^2]+2\,\mathrm{E}[(\mathrm{E}[\hat{f}(x_0)]-\hat{f}(x_0))(f(x_0)-E[\hat{f}(x_0)])]$$

Once more, ϵ is a random error term that is independent of x and has mean zero, so

$$\begin{aligned}
\mathrm{E}[\hat{f}(x_0)] &= \mathrm{E}[f(x_0) + \epsilon] \\
&= \mathrm{E}[f(x_0)] + \mathrm{E}[\epsilon] \\
&= f(x_0)
\end{aligned}$$

So we can infer that $2\,\mathrm{E}[(\mathrm{E}[\hat{f}(x_0)] - \hat{f}(x_0))(f(x_0) - E[\hat{f}(x_0)])] = 0$.
According to the property of variance and the definition of bias,

$$\mathrm{var}(\hat{f}(x_0)) = \mathrm{E}[(\mathrm{E}[\hat{f}(x_0)] - \hat{f}(x_0))^2], b_f(\hat{f}(x_0))^2 = (f(x_0) - E[\hat{f}(x_0)])^2 = E[(f(x_0) - E[\hat{f}(x_0)])^2]$$

So the equation could be written by

$$\begin{aligned}
\mathrm{E}[(Y_0 - \hat{f}(x_0))^2] &= \mathrm{E}[(f(x_0) - \hat{f}(x_0)^2] + \mathrm{var}(\epsilon) \\
&= \mathrm{var}(\hat{f}(x_0)) + b_f(\hat{f}(x_0))^2 + \mathrm{var}(\epsilon)
\end{aligned}$$